

Unit 2 Notes

Logic

Propositions, predicates,
reasoning, and proof

These notes develop the ideas from their basic meanings. Each section connects ordinary language to precise notation, explains a method, and includes a question with a worked solution. Diagrams use black outlines on white backgrounds so that the document can be printed without colour.

The coverage follows the supplied syllabus: prerequisite propositional logic, satisfiability, tautology, contradiction, validity, normal forms, predicates and quantifiers, nested quantifiers, rules of inference, proof techniques, mathematical induction, and answer extraction.

There are 52 numbered teaching sections. Read the explanation first, attempt each practice question before consulting its solution, and explain the reasoning in your own words afterward.

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1 How to read a logical statement

Logic studies what follows from what. Its central task is not to decide whether a sentence sounds plausible. It is to identify exactly what the sentence asserts and whether a proposed conclusion is forced by the assumptions.

Suppose a rule says, “Every registered student may enter the library.” If Mira is registered, the rule supports the conclusion that Mira may enter. If Mira may enter, however, we cannot conclude that she is registered: staff members might also have permission. This difference between a rule and its reversal appears throughout the unit.

Main idea. Separate three things: the *statement*, the *assumptions*, and the *reasoning*. A statement can be true even when someone gives a poor argument for it. An argument can have a valid structure even when one of its assumptions is false.

We use letters such as p and q for whole statements. Later, expressions such as $P(x)$ describe properties of an object x . These are different levels of description: p has a truth value once its meaning is fixed; $P(x)$ may still require a value for x .

Worked example. Let p mean “the file is saved” and q mean “the backup exists.” The sentence “If the file is saved, then the backup exists” is written $p \rightarrow q$. The arrow records a conditional relationship. It does not, by itself, assert p or assert q .

Practice. A rule says, “Every square is a rectangle.” A shape is a rectangle. Does the rule establish that it is a square?

Solution. No. A rectangle with side lengths 2 and 5 satisfies the description “rectangle” but is not a square. The rule moves from being a square to being a rectangle. It does not move in the reverse direction. The example also shows how a single well-chosen counterexample can expose an invalid inference.

2 Propositions and truth values

A *proposition* is a declarative statement that has a definite truth value: true or false. We work in classical two-valued logic, so a proposition has exactly one of these values in a fixed interpretation.

“Seven is prime” is a proposition. “Close the door” is an instruction and has no truth value. “Is the door closed?” is a question. “ $x > 3$ ” is not yet a proposition if x has not been assigned a value or bound by a quantifier.

Context matters. “It is raining” becomes definite only after the relevant place and time are understood. Unknown truth is different from absent truth: a historical statement may be a proposition even if we do not know whether it is true.

Main idea. An *atomic proposition* is treated as a single unit. A *compound proposition* combines statements with connectives such as “and,” “or,” and “not.” Atomic does not mean short; it means that we are not analysing its internal logical structure at the chosen level.

Worked example. Let p be “8 is even” and q be “8 is prime.” Then p is true and q is false. The compound statement “8 is even and prime” is false because its second requirement fails.

Practice. Classify: (a) 12 is divisible by 3; (b) find 12/3; (c) n is divisible by 3; (d) every integer divisible by 6 is divisible by 3.

Solution. (a) is a true proposition. (b) is a request, not a proposition. (c) is an open statement until the meaning of n is supplied. (d) is a true proposition: the word “every” binds the variable implicitly. If an integer equals $6k$, it also equals $3(2k)$, establishing the claimed divisibility.

Common error. Do not classify a sentence as “not a proposition” merely because proving it would be difficult. Logical status and difficulty are separate issues.

3 Negation and conjunction

The negation of p is written $\neg p$ and read “not p .” It reverses the truth value. The conjunction $p \wedge q$ means “ p and q ” and is true only when both statements are true.

p	q	$\neg p$	$p \wedge q$
T	T	F	T
T	F	F	F
F	T	T	F
F	F	T	F

Conjunction collects requirements. If access requires a valid card *and* a correct PIN, satisfying only one requirement is insufficient. In mathematics, “ x is positive and even” similarly imposes two independent conditions.

Main idea. Negation must deny the entire assertion, not replace it with a stronger assertion. The negation of “the score is greater than 50” is “the score is at most 50.” Saying “the score is less than 50” incorrectly excludes the boundary case 50.

Worked example. If p is true and q is false, then $p \wedge \neg q$ is true: the first part is true and the negation of the false second statement is also true. But $\neg(p \wedge q)$ is also true for a different structural reason: the conjunction inside the parentheses is false.

Practice. Let p mean $x \geq 2$ and q mean $x < 7$. Translate $p \wedge q$ and $\neg p$, assuming x is real.

Solution. The conjunction means $2 \leq x < 7$: both inequalities must hold at once. The negation of p is $x < 2$. Equality belongs to the original statement and therefore does not belong to its negation. At $x = 2$, the conjunction is true; at $x = 7$, it is false.

4 Inclusive and exclusive disjunction

The disjunction $p \vee q$ means “ p or q , or both.” It is false only when both alternatives are false. This inclusive meaning is the default mathematical interpretation of “or.”

Exclusive or, written here as $p \oplus q$, means that exactly one alternative is true. It can be expressed using the basic connectives:

$$p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q).$$

The symbol $\equiv$ means that the expressions have the same truth value in every assignment. The first part requires at least one true alternative; the second excludes the case in which both are true.

p	q	$p \vee q$	$p \oplus q$
T	T	T	F
T	F	T	T
F	T	T	T
F	F	F	F

Worked example. “An integer is divisible by 2 or by 3” includes 6, because 6 is divisible by both. If exactly one divisibility condition is intended, that qualification must be stated.

Practice. A form accepts a passport or a driving licence. A person supplies both. Does $p \vee q$ describe a rule that accepts this submission? Does $p \oplus q$?

Solution. The inclusive rule accepts it because at least one accepted document has been supplied. The exclusive rule rejects it because both conditions are true. The two models differ in exactly one truth-table row, but that row changes the practical meaning of the policy.

Common error. Everyday language sometimes uses “or” exclusively. Translate the intended condition rather than assuming every English “or” has the same interpretation.

5 Implication and the meaning of a conditional

The implication $p \rightarrow q$ says, “If p , then q .” Here p is the *antecedent* and q is the *consequent*. The conditional forbids one situation: p being true while q is false.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Why are the last two rows true? Consider the rule “If an application is approved, a confirmation is issued.” A rejected application does not violate this particular rule, regardless of whether some other message is issued. The rule makes a demand only in the approved case.

This is *material implication*. It does not automatically express causation, timing, or relevance. Additional assumptions are needed if those ideas matter.

Main idea. The equivalent form

$$p \rightarrow q \equiv \neg p \vee q$$

is useful for calculations. Either the triggering condition does not occur, or the required consequence holds.

Practice. Negate “If the battery is charged, the device starts.” Use p for “charged” and q for “starts.”

Solution. A failed conditional requires the trigger and a failed consequence, so the negation is $p \wedge \neg q$. In words, “The battery is charged and the device does not start.” Merely saying “the device does not start” is insufficient: perhaps the battery is not charged, in which case the original conditional has not been violated.

6 Converse, contrapositive, and biconditional

For $p \rightarrow q$, the *converse* is $q \rightarrow p$, the *inverse* is $\neg p \rightarrow \neg q$, and the *contrapositive* is $\neg q \rightarrow \neg p$. The original statement is equivalent to its contrapositive. The converse and inverse are equivalent to each other, but usually not to the original.

The biconditional $p \leftrightarrow q$ means “ p if and only if q .” It requires both directions:

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p).$$

It is true when p and q have the same truth value, including when both are false.

Main idea. In $p \rightarrow q$, p is *sufficient* for q , and q is *necessary* for p . “ p only if q ” therefore means $p \rightarrow q$. “ p if q ” means $q \rightarrow p$.

Worked example. “An integer is divisible by 4 only if it is even” means $D_4 \rightarrow E$. Divisibility by 4 is enough to ensure evenness; evenness is required for divisibility by 4. The converse fails at 6.

Practice. Translate “Access is granted if and only if the account is active.” Then state its two separate obligations.

Solution. With a for access and b for an active account, the translation is $a \leftrightarrow b$. The obligations are $a \rightarrow b$ and $b \rightarrow a$. The first prohibits access for an inactive account. The second prohibits denying access to an active account. A rule that enforces only the first condition is weaker than the biconditional.

Common error. When proving an “if and only if” theorem, label and prove both directions. A correct proof of one implication completes only half the task.

7 Constructing a truth table systematically

A truth table lists every possible assignment of truth values to the atomic propositions. With n distinct atoms there are 2^n rows, because each atom independently has two choices. Repeated occurrences of the same atom do not create new choices.

Use a fixed row order. For three atoms, let p stay true for four rows, let q change every two rows, and let r change every row. This prevents missing or duplicating an assignment. Evaluate inside parentheses first and create intermediate columns whenever the expression becomes hard to read.

Worked example. Examine $(p \rightarrow q) \wedge p$:

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	F

The final column is true only in the first row. In that row q is also true. This observation is the truth-table basis of modus ponens, a rule introduced later.

Practice. How many rows are needed for $(p \wedge q) \vee (\neg p \wedge r)$? Evaluate it when $p = F$, $q = F$, and $r = T$.

Solution. There are three distinct atoms, so eight rows are needed. At the specified assignment, $p \wedge q$ is false, while $\neg p \wedge r$ is true. Their disjunction is true. Notice that q is irrelevant on this particular branch because p is false; nevertheless, all three atoms still belong in the full truth table.

Common error. A partially completed table can prove satisfiability if it finds a true row. It cannot prove tautology unless every possible row has been checked or another complete argument is supplied.

8 Equivalence and the basic algebra of logic

Two formulas are *logically equivalent* if they agree under every truth assignment. We write $A \equiv B$. This is a relationship between formulas. The formula $A \leftrightarrow B$ is a tautology exactly when that relationship holds.

The following laws work for arbitrary formulas, not just single letters. Here T and F denote constant truth and falsehood.

Law	Equivalent forms	
Identity	$p \wedge T \equiv p$	$p \vee F \equiv p$
Domination	$p \vee T \equiv T$	$p \wedge F \equiv F$
Idempotence	$p \wedge p \equiv p$	$p \vee p \equiv p$
Double negation	$\neg\neg p \equiv p$	
Complement	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$

Conjunction and disjunction are commutative, so the order of their inputs can be exchanged. They are also associative, so a chain of the same connective can be regrouped without changing its value.

Worked example. Simplify $(p \wedge T) \vee (p \wedge F)$. The left term becomes p by identity. The right term becomes F by domination. The resulting $p \vee F$ is p .

Practice. Simplify $\neg\neg q \wedge (q \vee \neg q)$, naming the main steps.

Solution. Double negation changes the first factor to q . The complement law changes the second factor to T. Identity then gives $q \wedge T \equiv q$. Each replacement is valid inside the larger expression because equivalent subformulas are interchangeable.

Common error. Logical algebra resembles numerical algebra but has different laws. For example, $p \vee p \equiv p$ has no ordinary numerical analogue $x + x = x$. Use logical identities explicitly rather than transferring arithmetic habits without checking them.

9 De Morgan's laws and negating compound claims

De Morgan's laws explain how negation passes through “and” and “or”:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q,$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q.$$

The connective changes when the negation moves inward. To deny that both requirements hold, at least one must fail. To deny that either alternative holds, both must fail.

Worked example. Suppose successful login requires a valid password and an active account. The denial of successful satisfaction of these requirements is “the password is invalid or the account is inactive.” It need not be the case that both fail.

For three or more inputs, the same reasoning applies. The negation of $p \wedge q \wedge r$ is $\neg p \vee \neg q \vee \neg r$. Parentheses make the scope clear when different connectives occur together.

Practice. Simplify $\neg((p \vee q) \wedge \neg r)$ until negations occur only directly before atoms.

Solution. First apply De Morgan's law to the outer conjunction:

$$\neg(p \vee q) \vee \neg\neg r.$$

Next negate the inner disjunction and remove the double negation:

$$(\neg p \wedge \neg q) \vee r.$$

In words, the original requirement fails if both p and q fail, or if r holds. This explanation is useful for checking that the symbols retain the intended meaning.

Common error. Writing $\neg(p \vee q)$ as $\neg p \vee \neg q$ is wrong. If p is true and q is false, the first expression is false and the proposed replacement is true. A single disagreeing row disproves equivalence.

10 Distributive and absorption laws

Both connectives distribute over the other:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r),$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r).$$

The first law separates two ways of satisfying a shared requirement. The second law is just as valid, even though it feels less familiar from ordinary algebra.

Absorption removes a condition already covered by another:

$$p \vee (p \wedge q) \equiv p,$$

$$p \wedge (p \vee q) \equiv p.$$

For the first law, if p is true the whole expression is true; if p is false both alternatives are false. The value of q never changes the outcome.

Worked example. Simplify $(p \wedge q) \vee (p \wedge \neg q)$. Factor the common requirement p using distributivity:

$$p \wedge (q \vee \neg q) \equiv p \wedge \mathbf{T} \equiv p.$$

The expression describes two alternatives that together cover every possibility for q , while both require p .

Practice. Simplify $(p \vee q) \wedge (p \vee \neg q)$.

Solution. Use the second distributive law in reverse:

$$(p \vee q) \wedge (p \vee \neg q) \equiv p \vee (q \wedge \neg q)$$

$$\equiv p \vee \mathbf{F}$$

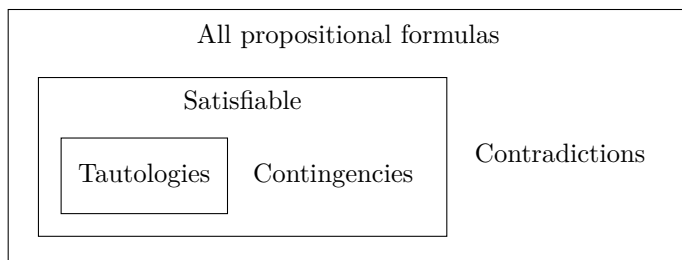
$$\equiv p.$$

To check the meaning directly, suppose p is false. Then the two parentheses demand both q and $\neg q$, which is impossible. If p is true, both parentheses hold regardless of q .

Common error. Distributing in the wrong direction can make a problem much longer. Decide whether your goal is simplification, a particular normal form, or comparison with another formula before expanding everything.

11 Satisfiability, tautology, and contradiction

A formula is *satisfiable* when at least one truth assignment makes it true. A satisfying assignment is also called a *model*. A *tautology* is true in every assignment. A *contradiction* is false in every assignment. A *contingent* formula is true in some assignments and false in others.



Every tautology is satisfiable, but many satisfiable formulas are not tautologies. A contradiction is also called *unsatisfiable*. These definitions concern all possible assignments, not just the situation currently under discussion.

Worked example. $p \vee \neg p$ is a tautology. $p \wedge \neg p$ is a contradiction. $p \rightarrow q$ is contingent: it is false at $p = \text{T}, q = \text{F}$ and true in the other rows.

Practice. Classify $(p \vee q) \wedge \neg p$. Give evidence for your classification.

Solution. It is contingent. At $p = \text{F}, q = \text{T}$, both the disjunction and $\neg p$ are true. At $p = \text{T}, q = \text{T}$, the second factor is false. One true row establishes satisfiability; one false row rules out tautology. Together they establish contingency.

Common error. “Not a tautology” does not mean “a contradiction.” There is a large middle category: formulas whose values depend on the assignment.

12 Validity of arguments and semantic entailment

An argument consists of premises and a conclusion. It is *valid* when no interpretation makes every premise true and the conclusion false. Validity is therefore about preservation of truth, not about whether the premises happen to describe reality.

For premises $P_1, \dots, P_k$ and conclusion C , write

$$P_1, \dots, P_k \models C.$$

The symbol $\models$ means “semantically entails.” The argument is valid exactly when $(P_1 \wedge \dots \wedge P_k) \rightarrow C$ is a tautology. Equivalently, the premises together with $\neg C$ are unsatisfiable.

Worked example. The premises $p \rightarrow q$ and p entail q . Any assignment making p true must make q true if the conditional is also to remain true. There is no row with both premises true and the conclusion false.

An argument is *sound* when it is valid and its premises are true. Soundness therefore adds a factual requirement to a structural one.

Practice. Are the premises $p \rightarrow q$ and q enough to entail p ?

Solution. No. Choose $p = \text{F}$ and $q = \text{T}$. The conditional is true because its antecedent is false, and the second premise q is true. The conclusion p is false. This is a countermodel, so the argument is invalid. It is unnecessary to inspect any further rows once such a countermodel is found.

Common error. If the premises are inconsistent, they entail every conclusion in classical logic: there is no model in which all the premises hold. Such an argument is valid in the formal sense, but it cannot be sound. Always distinguish validity from the usefulness or consistency of the assumptions.

13 Finding models and countermodels efficiently

A truth table is reliable but grows exponentially. For a small formula, direct reasoning can often find a model more quickly. Start with the strongest requirements, propagate their consequences, and check that all original conditions are satisfied.

Worked example. Consider

$$(p \vee q) \wedge (\neg p \vee r) \wedge \neg q.$$

The last requirement forces $q = \text{F}$. The first then forces $p = \text{T}$. The second now forces $r = \text{T}$. Checking the three original factors gives true, true, and true. Thus $(p, q, r) = (\text{T}, \text{F}, \text{T})$ is a model.

If the additional requirement $\neg r$ is imposed, the same chain forces both r and $\neg r$. The extended set is unsatisfiable. This is the basic intuition behind unit propagation in satisfiability procedures.

To refute an argument, search for a model of its premises *and the negation of its conclusion*. This combines the tasks of satisfying assumptions and violating the proposed result.

Practice. Do $p \vee q$ and $p \rightarrow r$ entail r ? If not, give a complete countermodel.

Solution. They do not. To falsify the conclusion, set $r = \text{F}$. Then the conditional forces $p = \text{F}$. To keep $p \vee q$ true, set $q = \text{T}$. Both premises are now true while r is false. The missing possibility is that the first premise is satisfied through q , in which case the rule about p need never be triggered.

Common error. A proposed model must specify enough values to evaluate every relevant atom. Also distinguish a contradiction reached on one guessed branch from a contradiction forced on every branch. Only the latter establishes unsatisfiability of the whole formula.

14 Literals, clauses, and normal forms

A *literal* is an atom or its negation: p and $\neg p$ are literals, but $\neg(p \vee q)$ is not. A *clause* is a disjunction of literals, such as $p \vee \neg q \vee r$. A *term*, in the propositional normal-form discussion, is a conjunction of literals.

A formula is in *conjunctive normal form* (CNF) if it is a conjunction of clauses. It is in *disjunctive normal form* (DNF) if it is a disjunction of terms. The outside connective supplies the name.

$$(p \vee \neg q) \wedge (q \vee r) \quad \text{is CNF,}$$

$$(p \wedge \neg q) \vee (q \wedge r) \quad \text{is DNF.}$$

A single literal can count as a one-literal clause or term. Thus $p \wedge q$ is both CNF, as a conjunction of singleton clauses, and DNF, as one conjunctive term.

Main idea. CNF describes simultaneous constraints: every clause must be satisfied. DNF describes alternative successful cases: at least one term must be satisfied. This difference explains why each form is useful for different tasks.

Worked example. $(p \rightarrow q) \wedge r$ is not yet in CNF because an implication remains. Replacing the conditional with $\neg p \vee q$ gives $(\neg p \vee q) \wedge r$, which is CNF.

Practice. Classify $(p \wedge q) \vee \neg r$ and $\neg(p \wedge q) \vee r$ as written.

Solution. The first is DNF: it is a disjunction of the term $p \wedge q$ and the singleton term $\neg r$. The second is neither of the stated normal forms as written, because a negation covers a compound expression. After De Morgan's law it becomes $\neg p \vee \neg q \vee r$, a single clause and hence CNF. It is also DNF if each literal is viewed as a singleton term.

15 Converting a formula into CNF

To obtain CNF, remove biconditionals and implications, move negations inward using De Morgan's laws, eliminate double negations, and distribute disjunction over conjunction. Simplify only with valid equivalence laws.

The intermediate stage where negations occur only on atoms and the remaining connectives are $\wedge$ and $\vee$ is called *negation normal form*. It is a useful checkpoint, but it need not already be CNF or DNF.

Worked example. Convert $p \rightarrow (q \wedge r)$ to CNF. First remove the arrow:

$$\neg p \vee (q \wedge r).$$

Then distribute the disjunction:

$$(\neg p \vee q) \wedge (\neg p \vee r).$$

The original conditional requires both q and r when p holds. The result expresses these as two separate constraints, each switched off when p is false.

Practice. Convert $\neg(p \vee q) \vee (r \wedge s)$ to CNF.

Solution. De Morgan's law gives $(\neg p \wedge \neg q) \vee (r \wedge s)$. Distributing over the second conjunction first produces

$$((\neg p \wedge \neg q) \vee r) \wedge ((\neg p \wedge \neg q) \vee s).$$

Distribute once more in each parenthesis:

$$\begin{aligned} &(\neg p \vee r) \wedge (\neg q \vee r) \\ &\wedge (\neg p \vee s) \wedge (\neg q \vee s). \end{aligned}$$

Each component is now a clause. Four clauses are needed by this straightforward expansion because every literal in the first conjunction must be paired with every requirement in the second.

Common error. Conversion can increase size substantially. CNF is a standard shape, not necessarily the shortest expression. Repeated distribution may cause exponential growth for some families of formulas.

16 Converting a formula into DNF

DNF lists sufficient combinations of literal conditions. After removing arrows and moving negations inward, distribute conjunction over disjunction. Each resulting term should contain only literals joined by $\wedge$.

Worked example. Convert $(p \vee q) \wedge \neg r$ into DNF:

$$(p \wedge \neg r) \vee (q \wedge \neg r).$$

There are two ways to meet the original requirements: p together with $\neg r$, or q together with $\neg r$. Both terms may be true simultaneously; ordinary DNF does not require disjoint alternatives.

Practice. Find and simplify a DNF for $(p \vee q) \wedge (\neg p \vee r)$.

Solution. Distribute to get

$$(p \wedge \neg p) \vee (p \wedge r) \vee (q \wedge \neg p) \vee (q \wedge r).$$

The first term is impossible, so it can be removed. A valid DNF is therefore

$$(p \wedge r) \vee (\neg p \wedge q) \vee (q \wedge r).$$

The last term is redundant. Whenever $q \wedge r$ holds, either p is true, making $p \wedge r$ true, or p is false, making $\neg p \wedge q$ true. Thus a shorter equivalent DNF is

$$(p \wedge r) \vee (\neg p \wedge q).$$

This is the consensus principle in a concrete form. The explanation matters more than memorising an additional identity.

Common error. Do not remove a term merely because it shares a literal with another term. A removal is justified only when all assignments covered by that term are already covered elsewhere. Testing a few examples can reveal a bad simplification, but proving redundancy requires an argument covering every case.

17 Canonical forms from a truth table

A *minterm* contains every selected variable exactly once, either positive or negated, joined by conjunction. It is true in exactly one row. For the row $p = T, q = F, r = T$, the minterm is $p \wedge \neg q \wedge r$.

A *maxterm* contains every variable exactly once joined by disjunction and is false in exactly one row. For that same row, the maxterm is $\neg p \vee q \vee \neg r$. Notice the reversed signs: each literal must be false in the selected row.

The *principal DNF* is the disjunction of minterms for all true rows. The *principal CNF* is the conjunction of maxterms for all false rows. These are also called canonical DNF and canonical CNF, relative to the declared variable list.

Worked example. Exclusive or is true at (T, F) and (F, T), so its principal DNF is

$$(p \wedge \neg q) \vee (\neg p \wedge q).$$

Its false rows are (T, T) and (F, F), giving principal CNF

$$(\neg p \vee \neg q) \wedge (p \vee q).$$

Practice. Find both canonical forms of $p \rightarrow q$, using variables p, q .

Solution. The only false row is (T, F), giving principal CNF $\neg p \vee q$. The three true rows give principal DNF

$$(p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q).$$

Every variable occurs in every minterm. The shorter equivalent formula $\neg p \vee q$ is a DNF under the broad definition, but it is not a principal DNF over these two variables.

Common error. A minterm matches a row; a maxterm excludes a row. Keeping these purposes in mind prevents sign errors better than memorising two unrelated recipes.

18 Review: propositional reasoning in one problem

Consider a system with three statements: p means “permission is granted,” q means “the account is verified,” and r means “a supervisor approves.” The policy is

$$F = (p \rightarrow q) \wedge (p \rightarrow r) \wedge ((q \wedge r) \rightarrow p).$$

Read it before manipulating it. Permission requires verification and approval. Conversely, having both verification and approval is enough for permission. Together, the three requirements say that permission is granted exactly when both conditions hold.

Worked example. In compact form, $F \equiv p \leftrightarrow (q \wedge r)$. Eliminating the arrows gives the CNF

$$(\neg p \vee q) \wedge (\neg p \vee r) \wedge (\neg q \vee \neg r \vee p).$$

This form makes each prohibited combination explicit. The first clause rules out permission without verification, the second permission without approval, and the third denial of permission when both are present.

Practice. (a) Is F satisfiable? (b) Is F a tautology? (c) Do F and q entail p ? (d) Do F, q, r entail p ?

Solution. (a) Yes. The assignment $p = q = r = \text{T}$ satisfies every clause. (b) No. The assignment $p = \text{T}, q = \text{F}, r = \text{T}$ violates the first clause. Hence F is contingent.

(c) No. Set $q = \text{T}, r = \text{F}, p = \text{F}$. The policy holds, and q holds, but permission is false because approval is missing. (d) Yes. The third implication now has a true antecedent, so p must be true.

This example combines translation, equivalence, normal forms, models, and validity. In a longer question, handle these as separate steps. A correct translation gives the later algebra a meaningful target.

19 Predicates, domains, and open statements

Propositional letters hide internal structure. Predicate logic exposes it. A *predicate* describes a property or relation, and its arguments specify the objects being discussed. For example, $E(x)$ may mean “ x is even,” while $L(x, y)$ may mean “ x is less than y .”

The *domain of discourse* is the collection of allowed objects. Unless stated otherwise, first-order domains in these notes are nonempty. A statement about all integers may have a different truth value when its domain changes to real numbers.

Worked example. Let $P(x)$ mean $x^2 = 4$. Then $P(2)$ and $P(-2)$ are true, while $P(3)$ is false. The open expression $P(x)$ is not assigned a single truth value until x receives a value or becomes quantified.

A constant, such as a , names an object. A function symbol, such as $f(x)$, denotes an object obtained from another object. A predicate symbol produces a truth-valued assertion. Thus $f(x)$ and $P(x)$ play different grammatical roles even when their notation looks similar.

Practice. Let the domain be integers and $D(x, y)$ mean “ x divides y .” Interpret $D(3, 12)$, $D(12, 3)$, and $D(x, 12)$.

Solution. $D(3, 12)$ is true because $12 = 3 \cdot 4$. $D(12, 3)$ is false because no integer multiplied by 12 equals 3. The expression $D(x, 12)$ is an open statement specifying a condition on x ; it becomes true for integer divisors of 12. Argument order is essential because divisibility is not symmetric.

Common error. Always declare the domain and meanings of predicates before translating a substantial sentence. Symbols without these declarations can conceal ambiguity rather than remove it.

20 Universal quantification

The statement $\forall x P(x)$ means that $P(x)$ holds for every object in the declared domain. The symbol $\forall$ is the universal quantifier. The variable x is bound by it, and the formula following the quantifier is its scope.

To establish a universal claim, reason about an arbitrary domain element without relying on special properties not shared by all elements. To refute it, provide one domain element for which the predicate fails. Such an element is a *counterexample*.

Worked example. Over integers, $\forall x (x^2 \geq 0)$ is true. Over integers, $\forall x (x^2 > x)$ is false: choose $x = 0$ or $x = 1$. Testing $x = 2, 3, 4$ would not have justified the universal statement.

On a finite domain $\{a, b, c\}$, universal quantification is equivalent to a conjunction:

$$\forall x P(x) \equiv P(a) \wedge P(b) \wedge P(c).$$

On an infinite domain, a finite sample cannot replace the quantifier.

Practice. Translate “Every student in the class submitted the assignment.” First use a domain consisting only of class members, then use a domain of all people.

Solution. On the restricted domain, write $\forall x S(x)$, where $S(x)$ means that x submitted. On the larger domain, let $C(x)$ mean membership in the class and write

$$\forall x (C(x) \rightarrow S(x)).$$

The implication limits the requirement to class members. Replacing it with $C(x) \wedge S(x)$ would claim that every person belongs to the class as well as having submitted, which is much stronger.

Common error. A universal claim over a category with no members can be true vacuously. “Every absent student submitted” imposes no violated condition if there are no absent students. It does not establish that an absent student exists.

21 Existential quantification and witnesses

The statement $\exists x P(x)$ means that at least one domain element satisfies P . The symbol $\exists$ is the existential quantifier. An element making the predicate true is called a *witness*.

An existential statement does not normally claim uniqueness. “There exists a solution” allows two, ten, or infinitely many solutions. If exactly one is intended, existence and uniqueness must both be stated.

Worked example. Over integers, $\exists x (x^2 = 9)$ is true because $x = 3$ is a witness. The value -3 is another witness. Over integers, $\exists x (x^2 = 2)$ is false; over reals, it is true. The quantifier cannot be interpreted independently of the domain.

On the finite domain $\{a, b, c\}$,

$$\exists x P(x) \equiv P(a) \vee P(b) \vee P(c).$$

This is the existential counterpart of the universal conjunction.

Practice. Translate “Some student passed every required examination,” using people as the domain for x and examinations as the domain for y . Let $S(x)$ mean student, $R(y)$ mean required, and $P(x, y)$ mean passed.

Solution. Write

$$\exists x (S(x) \wedge \forall y (R(y) \rightarrow P(x, y))).$$

The conjunction ensures that the witness really is a student. The inner implication makes a passing requirement only for required examinations. Using $S(x) \rightarrow \dots$ at the outer level would allow a nonstudent to satisfy the existential statement merely by making its antecedent false.

Common error. When proving existence, show both that the proposed witness belongs to the domain and that it satisfies the whole required property. Naming an object without checking its conditions is an incomplete proof.

22 Negating quantified statements

To deny “every object has the property,” it is enough to find one object without it. To deny “some object has the property,” every object must lack it. Therefore

$$\neg \forall x P(x) \equiv \exists x \neg P(x),$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x).$$

These are the quantifier versions of De Morgan’s laws. Each quantifier changes when a negation passes through it.

Worked example. Negate “Every student passed at least one test.” A formal version is $\forall x \exists y P(x, y)$ when the domains already consist of students and tests. Moving the negation inward gives

$$\exists x \forall y \neg P(x, y).$$

In words, “There is a student who passed no test.” This is stronger than saying that some student failed some test; the original statement permits failed tests as long as each student passed at least one.

Practice. Over real numbers, negate $\forall x \exists y (y > x \wedge y < x + 1)$.

Solution. First switch the universal quantifier, then the existential one, then negate the conjunction:

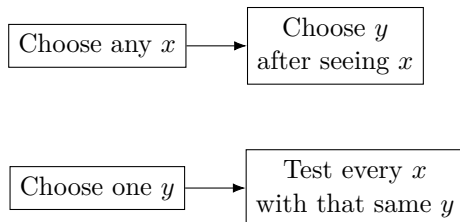
$$\exists x \forall y (y \leq x \vee y \geq x + 1).$$

The result says that some open interval $(x, x + 1)$ contains no real number. It is false, since $y = x + \frac{1}{2}$ lies in every such interval. The negation’s falsity is consistent with the original statement’s truth.

Common error. After switching quantifiers, remember to negate the predicate itself. Also preserve boundary cases: the negation of $y > x$ is $y \leq x$, not $y < x$.

23 Nested quantifiers and order of choice

Mixed quantifiers have an order that controls dependence. In $\forall x \exists y R(x, y)$, the witness y may depend on x . In $\exists y \forall x R(x, y)$, one fixed y must work for every x .



Worked example. Over integers, $\forall x \exists y (y > x)$ is true: after receiving x , choose $y = x + 1$. But $\exists y \forall x (y > x)$ is false. For any fixed candidate y , the choice $x = y$ defeats the strict inequality.

Quantifiers of the same kind commute: $\forall x \forall y$ can be swapped, and $\exists x \exists y$ can be swapped. Mixed quantifiers cannot generally be swapped. The statement with a single uniform witness, $\exists y \forall x R(x, y)$, does imply $\forall x \exists y R(x, y)$, but the reverse implication need not hold.

Practice. Compare “Every book has an author” with “There is a person who authored every book.”

Solution. The first is $\forall b \exists a A(a, b)$; different books may have different authors. The second is $\exists a \forall b A(a, b)$; one person must qualify for every book. Two books with different sole authors provide a situation in which the first statement is true and the second false.

24 Relations as tables: seeing quantifier order

A finite relation can be displayed as a table. Rows represent objects chosen for the first variable and columns represent objects chosen for the second. A 1 means that the relation holds; a 0 means that it does not.

$R(x, y)$	$y = a$	$y = b$	$y = c$
$x = 1$	1	0	0
$x = 2$	0	1	0
$x = 3$	0	0	1

Here $\forall x \exists y R(x, y)$ is true because each row contains a 1. But $\exists y \forall x R(x, y)$ is false because no column consists entirely of 1s. The table gives a concrete counterexample to swapping mixed quantifiers.

The statement $\exists x \forall y R(x, y)$ asks for a row entirely of 1s. The statement $\forall y \exists x R(x, y)$ asks for at least one 1 in every column. Read the outer quantifier first, then translate the inner requirement into a row or column condition.

Practice. Replace the last row by $(1, 1, 1)$. Which of the four statements in this section become true?

Solution. Every row still has a 1, so $\forall x \exists y R(x, y)$ remains true. Every column has a 1 in the last row, so $\forall y \exists x R(x, y)$ is true. There is now an all-1 row, making $\exists x \forall y R(x, y)$ true. There is still no all-1 column because the first two rows have their 1s in different columns. Thus $\exists y \forall x R(x, y)$ remains false.

This visual method is especially useful when a verbal description is confusing. Construct the smallest finite example that displays the intended dependence, then read the quantifiers directly from it.

25 Scope, free variables, and careful substitution

A variable occurrence is *bound* when it lies within the scope of a matching quantifier. Otherwise it is *free*. A formula with no free variables is a sentence. The same variable name can have different roles in different parts of a formula, so parentheses matter.

Worked example. In $(\forall x P(x)) \vee Q(x)$, the occurrence inside $P(x)$ is bound, but the occurrence inside $Q(x)$ is free. In $\forall x(P(x) \vee Q(x))$, both are bound. These formulas need not have the same meaning.

Bound variable names are labels, not values. Renaming $\forall x P(x)$ to $\forall z P(z)$ changes no meaning if the new name does not interfere with another variable's scope.

Main idea. Substitution must avoid *variable capture*. If a free variable is replaced by a name already bound nearby, the replacement can accidentally change a parameter into a quantified variable.

Practice. In $\exists y L(x, y)$, replace the free variable x by y without capturing it.

Solution. First rename the bound variable to a fresh name, giving $\exists z L(x, z)$. Then substitute the free y for x , obtaining $\exists z L(y, z)$. Directly writing $\exists y L(y, y)$ would be wrong: it asks whether some object is related to itself, rather than whether the object specified by the free parameter y is related to something.

A related issue appears when distributing quantifiers. For example, $\forall x(P(x) \wedge Q(x))$ is equivalent to $(\forall x P(x)) \wedge (\forall x Q(x))$. But a universal quantifier does not generally distribute over disjunction. On a two-element domain, one object may satisfy only P and the other only Q .

Common error. Before replacing or moving a variable, identify its scope. Many apparently algebraic errors in predicate logic are actually changes in which variable occurrences are bound.

26 Exactly one, at least two, and restricted claims

Some statements require more than a single quantifier. “Exactly one object has property P ” combines existence with uniqueness:

$$\exists x(P(x) \wedge \forall y(P(y) \rightarrow y = x)).$$

The witness x has the property, and every object with the property must equal that witness. Equality here means identity of objects, not merely a shared property.

“At least two objects have P ” is

$$\exists x \exists y (P(x) \wedge P(y) \wedge x \neq y).$$

The inequality is necessary: two variable names can denote the same object. “At most one” can be expressed by $\forall x \forall y ((P(x) \wedge P(y)) \rightarrow x = y)$; this permits no objects with the property.

Worked example. Let $C(x)$ mean that x is a course and $T(y, x)$ mean that person y teaches it. “Every course has exactly one teacher” becomes

$$\forall x (C(x) \rightarrow \exists y (T(y, x) \wedge \forall z (T(z, x) \rightarrow z = y))).$$

The teacher may vary from course to course because the existential quantifier is inside the universal one.

Practice. Translate “There are exactly two integers whose square is 1.”

Solution. Over integers, write

$$\begin{aligned} \exists x \exists y (x^2 = 1 \wedge y^2 = 1 \wedge x \neq y \\ \wedge \forall z (z^2 = 1 \rightarrow (z = x \vee z = y))). \end{aligned}$$

The witnesses are 1 and -1 . The final universal condition excludes a third distinct solution. Existence alone would not do this. To justify the mathematics, factor $z^2 - 1 = (z - 1)(z + 1)$; a product of integers is zero only if one factor is zero.

27 Rules of propositional inference

A rule of inference is a small valid argument pattern that can be reused inside a larger proof. In the patterns below, p, q, r may stand for compound formulas as well as atoms.

Rule	Premises	Conclusion
Modus ponens	$p \rightarrow q, p$	q
Modus tollens	$p \rightarrow q, \neg q$	$\neg p$
Hypothetical syllogism	$p \rightarrow q, q \rightarrow r$	$p \rightarrow r$
Disjunctive syllogism	$p \vee q, \neg p$	q
Conjunction	p, q	$p \wedge q$
Simplification	$p \wedge q$	p
Addition	p	$p \vee q$

Modus ponens applies a rule when its trigger is known. Modus tollens uses a failed consequence to rule out its trigger. Addition may seem to introduce unnecessary information, but the conclusion is weaker than the premise: if p is true, at least one of p, q is certainly true.

Worked example. From $p \rightarrow q, q \rightarrow r$, and $\neg r$, first obtain $\neg q$ by modus tollens on the second implication. Then obtain $\neg p$ by modus tollens on the first. Each step uses a previously available statement.

Practice. Derive s from $p \vee q, \neg p$, and $q \rightarrow s$.

Solution. The first two premises yield q by disjunctive syllogism. Combining q with $q \rightarrow s$ yields s by modus ponens. A clear formal proof would number the premises and cite their numbers beside each derived line. This records which information supports which conclusion.

Common error. A rule applies to a complete matching pattern. From $p \vee q$ alone, neither p nor q may be selected. An alternative is known to hold, but the premise does not say which one.

28 Invalid inference patterns and counterexamples

Two frequent errors resemble valid rules but reverse the direction of the information.

Affirming the consequent uses $p \rightarrow q$ and q to infer p . It is invalid because q may hold for another reason. *Denying the antecedent* uses $p \rightarrow q$ and $\neg p$ to infer $\neg q$. It is invalid because the conditional does not state that p is the only route to q .

Worked example. “If a number is divisible by 4, it is even. The number is even. Therefore it is divisible by 4.” The number 6 defeats the argument. For denying the antecedent, use “6 is not divisible by 4, therefore it is not even.” The same number again exposes the error.

The assignment $p = \text{F}, q = \text{T}$ is a countermodel to both patterns. In one pattern it makes the asserted p false; in the other it makes the asserted $\neg q$ false, while the premises remain true.

Practice. A student argues: “If the program is correct, it passes the test. It passes the test, so it is correct.” Identify the pattern and repair the conclusion without adding unsupported information.

Solution. This affirms the consequent. A program with an untested defect could pass the particular test. From the premises alone, correctness is not established. A justified statement is simply that passing this test is consistent with correctness. If the test had been proved sufficient for correctness, that would supply the missing converse, but it is an additional premise.

Common error. An invalid pattern can sometimes have a true conclusion. The issue is that the conclusion is not guaranteed by the given premises. Never use the accidental truth of one example to defend the general validity of an argument form.

29 Inference rules for quantified statements

Universal instantiation allows $P(t)$ to be inferred from $\forall xP(x)$, provided t denotes an appropriate domain object and substitution is free from variable capture. Existential generalisation allows $\exists xP(x)$ to be inferred from a particular instance $P(t)$.

Universal generalisation goes in the other direction only with a restriction: to conclude $\forall xP(x)$ from a proof of $P(a)$, the object a must be arbitrary, not selected because it has a special property. Assumptions on which the proof depends must not restrict a in an impermissible way.

Existential instantiation introduces a *fresh* temporary name for a witness. From $\exists xP(x)$, reason with $P(c)$ for a new name c . You may not assume that this witness is a previously named object, and the final conclusion must not depend on its arbitrary name.

Worked example. From $\forall x(S(x) \rightarrow M(x))$ and $S(a)$, instantiate the universal premise to get $S(a) \rightarrow M(a)$, then use modus ponens to obtain $M(a)$. Finally, infer $\exists xM(x)$ if an existential conclusion is desired.

Practice. From $\exists x(S(x) \wedge P(x))$ and $\forall x(P(x) \rightarrow Q(x))$, prove $\exists x(S(x) \wedge Q(x))$.

Solution. Let c be a fresh witness with $S(c) \wedge P(c)$. Simplification gives $S(c)$ and $P(c)$. Instantiating the universal premise gives $P(c) \rightarrow Q(c)$, so $Q(c)$ follows. Conjoin $S(c)$ and $Q(c)$ and then existentially generalise. The final statement does not mention c , so it does not depend on which witness received that temporary name.

Common error. From $\exists xP(x)$ and $\exists xQ(x)$, one cannot generally infer $\exists x(P(x) \wedge Q(x))$. The two existential statements may have different witnesses.

30 What a mathematical proof must establish

A proof is a finite chain of justified reasoning from stated assumptions, definitions, and previously established results to a precise conclusion. Its aim is to account for every permitted case, even when the domain is infinite.

Begin by writing the claim's logical structure. Is it universal, existential, conditional, or biconditional? Then expand key definitions. For integers, “even” means $2k$ for some integer k ; “odd” means $2k + 1$; and $a \mid b$ means $b = ak$ for some integer k .

Main idea. Proof writing has two stages. In exploration, work backward, try examples, and search for a useful representation. In the final proof, present the reasoning in an order that actually establishes the claim. An exploratory calculation is not automatically a valid final argument.

Worked example. To prove that a sum of two even integers is even, the useful representation is not their decimal digits. Write the integers as $2r$ and $2s$, where $r, s \in \mathbb{Z}$. Their sum is $2(r + s)$, and $r + s$ is an integer. That last fact connects the calculation back to the definition of evenness.

Practice. What is missing from the argument “The product of two odd numbers is odd because $3 \cdot 5 = 15$ and $5 \cdot 7 = 35$ ”?

Solution. The examples cover only two pairs. A proof must treat arbitrary odd integers. Write them as $2r + 1$ and $2s + 1$. Their product is $2(2rs + r + s) + 1$, which is odd because $2rs + r + s$ is an integer. This computation covers negative odd integers and all positive odd integers at once.

Common error. Avoid using the desired conclusion as an unproved step. If you begin by assuming an equation that you are supposed to establish, every transformation must be reversible and the final presentation must make the logical direction clear.

31 Direct proof from definitions

A direct proof of $P \rightarrow Q$ assumes P and derives Q . It is usually the first method to try when the hypothesis has a concrete algebraic or structural meaning.

Worked example. Prove that if $a \mid b$ and $b \mid c$ for integers a, b, c , then $a \mid c$. By the definition of divisibility, there are integers r, s such that $b = ar$ and $c = bs$. Substituting the first equality into the second gives

$$c = (ar)s = a(rs).$$

Since rs is an integer, this equality is exactly the definition of $a \mid c$. The proof does not need division by a , so it avoids unnecessary difficulties if $a = 0$.

This illustrates a reliable pattern: translate the hypotheses into usable equations, combine them, and identify the defining form of the conclusion. State why newly formed quantities still belong to the required domain.

Practice. Prove that the sum of an even integer and an odd integer is odd.

Solution. Let $a = 2r$ and $b = 2s + 1$, where $r, s \in \mathbb{Z}$. Then

$$a + b = 2r + 2s + 1 = 2(r + s) + 1.$$

Because $r + s \in \mathbb{Z}$, the sum has the defining form of an odd integer. The arbitrary choices of a and b establish the universal claim, rather than only a particular numerical example.

Common error. Use separate names for independent witnesses. Writing two arbitrary even integers as $2k$ and $2k$ would accidentally make them equal. Write $2r$ and $2s$ unless equality is part of the assumptions.

The concluding sentence of a proof should connect the last expression to the exact claim. A correct calculation can still leave the reader uncertain if the property it establishes is not identified.

32 Proof by contraposition

Because $P \rightarrow Q$ is equivalent to $\neg Q \rightarrow \neg P$, an implication may be proved through its contrapositive. This is useful when the negation of the conclusion gives a simpler description than the original hypothesis.

Worked example. Prove: if n^2 is even, then n is even, for $n \in \mathbb{Z}$. The direct hypothesis $n^2 = 2k$ does not immediately describe n . Instead prove: if n is odd, then n^2 is odd. Write $n = 2r + 1$. Then

$$n^2 = 4r^2 + 4r + 1 = 2(2r^2 + 2r) + 1.$$

The bracketed quantity is an integer, so n^2 is odd. This establishes the contrapositive and therefore the original implication.

We used the fact that every integer is either even or odd. Over another domain, the phrase “not even” may not automatically mean “odd,” so domain information remains part of the reasoning.

Practice. Prove that if $a + b$ is odd, then at least one of the integers a, b is odd.

Solution. Negate the conclusion carefully. The negation of “at least one is odd” is “both are even.” Assume $a = 2r$ and $b = 2s$. Then $a + b = 2(r + s)$ is even, hence not odd. This proves the contrapositive. It does not claim that every pair with an odd member has odd sum; two odd members have an even sum.

Common error. Contraposition changes both the order and the signs. The converse $Q \rightarrow P$ is not an acceptable replacement for $P \rightarrow Q$. Write the intended contrapositive explicitly before beginning the algebra.

33 Proof by contradiction

To prove a statement S by contradiction, assume $\neg S$ and derive an impossibility using valid reasoning. The contradiction might be an equation such as $0 = 1$, a statement together with its negation, or a violation of a known theorem.

For a conditional theorem $P \rightarrow Q$, its negation is $P \wedge \neg Q$. Thus a contradiction proof assumes the hypothesis and denies the conclusion. The original assumptions remain active throughout.

Worked example. Prove that $\sqrt{2}$ is irrational. Suppose instead that $\sqrt{2} = a/b$ for integers a, b in lowest terms with $b \neq 0$. Squaring gives $a^2 = 2b^2$. Thus a^2 is even, so a is even by the preceding section. Write $a = 2k$. Substitution gives $4k^2 = 2b^2$, hence $b^2 = 2k^2$, so b is also even. Both numerator and denominator are divisible by 2, contradicting the choice of lowest terms. Therefore no such rational representation exists.

Practice. Prove that there is no largest integer.

Solution. Suppose a largest integer M exists. The integer $M + 1$ is greater than M , contradicting its supposed maximality. Therefore the assumption of a largest integer is false. The proof uses closure of integers under adding 1 and the order property $M + 1 > M$.

Common error. A surprising consequence is not necessarily a contradiction. State the incompatible facts explicitly. Also distinguish contradiction from contraposition: contraposition proves another implication directly, while contradiction assumes the negation of the whole desired claim and rules it out.

Use contradiction when denial creates a useful object, such as a smallest counterexample, a rational representation, or a supposed extremal element. If a direct proof is shorter and clearer, there is no need to introduce an indirect argument.

34 Proof by cases

A proof by cases divides the possibilities into a collection that covers the whole domain, proves the conclusion in each case, and then combines the results. The cases need not always be disjoint, but they must be exhaustive.

Worked example. Prove that $n(n+1)$ is even for every integer n . If n is even, the product has an even factor. If n is odd, $n+1$ is even, so the product again has an even factor. Every integer is either even or odd, so these two cases complete the proof.

A more algebraic presentation writes $n = 2k$ in the first case and $n = 2k + 1$ in the second. Then the products become $2k(2k+1)$ and $(2k+1)2(k+1)$, both visibly divisible by 2.

Practice. Prove that n^2 leaves remainder 0 or 1 on division by 3.

Solution. Every integer has one of the forms $3k$, $3k+1$, or $3k+2$. Squaring gives

$$\begin{aligned}(3k)^2 &= 3(3k^2), \\ (3k+1)^2 &= 3(3k^2+2k)+1, \\ (3k+2)^2 &= 3(3k^2+4k+1)+1.\end{aligned}$$

Thus the remainder is 0 in the first case and 1 in the other two. The division algorithm guarantees that no integer lies outside these cases.

Common error. When dividing into positive and negative cases, remember zero. When dividing by a modulus m , include all remainders $0, 1, \dots, m-1$. An omitted boundary case can invalidate an otherwise correct proof.

Good case divisions arise from the structure of the problem. Parity suggests two cases; division by 3 suggests three; an absolute value suggests cases according to sign.

35 Biconditional, existence, and uniqueness proofs

A biconditional $P \leftrightarrow Q$ requires two implications. An existence-and-uniqueness theorem requires two different tasks: construct or establish a witness, then show that any other witness must equal it. Keeping the obligations separate makes omissions easy to detect.

Worked example. Prove that n is even if and only if n^2 is even. In the forward direction, write $n = 2k$, giving $n^2 = 2(2k^2)$. In the reverse direction, use the contrapositive proof that an odd integer has an odd square. The two directions use different techniques, which is entirely acceptable.

For uniqueness, the standard method is to suppose u and v both satisfy the defining conditions and derive $u = v$. This does not assume that two distinct solutions exist; it shows that two candidate solutions cannot be distinct.

Practice. Prove that for each real a , there is exactly one real x satisfying $2x + 3 = a$.

Solution. For existence, choose $x = (a-3)/2$. Substituting gives $2((a-3)/2) + 3 = a$, so the proposed real number works. For uniqueness, suppose u and v both satisfy the equation. Then $2u + 3 = a = 2v + 3$. Subtracting 3 and dividing by the nonzero number 2 gives $u = v$. Thus the solution exists and is unique.

Common error. Finding a solution proves existence, not uniqueness. Conversely, showing that any two solutions must agree does not prove that a solution exists. An equation with no solutions also satisfies the condition “any two solutions are equal” vacuously.

When a theorem includes “exactly,” “only,” or “if and only if,” underline those words during planning. They usually signal an additional proof obligation beyond a single construction or implication.

36 Counterexamples and the limits of examples

A universal statement is refuted by one counterexample. An existential statement is established by one witness. These are different logical uses of a single example. Neither task requires an extensive list once a correct example is found.

For a conditional universal claim, a counterexample must satisfy the hypothesis and fail the conclusion. A number outside the hypothesis says nothing about whether the implication is universally true.

Worked example. The statement “For every integer n , if n is odd then n is prime” is refuted by $n = 9$. It is odd and not prime. The value 8 is not a counterexample, because it does not satisfy the hypothesis.

To refute a claimed equivalence, it is enough to find an interpretation where one side is true and the other false. To refute an argument, satisfy every premise while falsifying the conclusion. These are three versions of the same underlying discipline: target the exact logical structure.

Practice. A conjecture says that $n^2 + n + 41$ is prime for every nonnegative integer n . Find a counterexample without trying many unrelated values.

Solution. Choose $n = 41$. Then

$$41^2 + 41 + 41 = 41(41 + 2) = 41 \cdot 43,$$

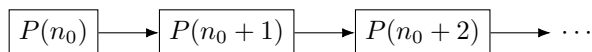
which is composite. The constant term suggests choosing a multiple of 41 so that all terms share the factor. This does not explain every value of the polynomial, but one permitted failure is enough to disprove the universal conjecture.

Common error. A conjecture surviving hundreds of checks has evidence in its favour, not a proof over an infinite domain. On a finite domain, exhaustive checking can be a proof, but the exhaustiveness itself must be established.

Counterexamples are also diagnostic tools. After finding one, ask which assumption could be added or which conclusion weakened to obtain a true and useful revised statement.

37 Mathematical induction: the logical structure

Induction proves a statement $P(n)$ for all integers $n \geq n_0$. It requires a base case $P(n_0)$ and an inductive step showing that, for an arbitrary $k \geq n_0$, $P(k)$ implies $P(k+1)$.



The base case starts the chain. The inductive step supplies every link. An unbroken chain without a starting point proves no particular instance, and a starting point without the links proves only the base case.

Main idea. The induction hypothesis is a temporary assumption used to prove a conditional. It is not an assumption that the entire theorem is already true. The induction principle then converts the base case and all successor implications into the universal conclusion.

Practice. A student verifies $P(1)$ and proves $P(k) \rightarrow P(k+2)$. Has the student proved $P(n)$ for every positive integer?

Solution. No. Starting at 1, the step reaches only 3, 5, 7, ... The even integers remain uncovered. A second base case $P(2)$ would start another chain reaching 4, 6, 8, ..., after which the two chains cover every positive integer. Alternatively, the student could prove a successor step of size 1.

Common error. Write the starting index explicitly. A theorem valid for $n \geq 4$ should not be forced into a base case at 1. Also check that the inductive step applies at the first transition; an inequality used in the step may require a larger k than expected.

38 Induction for a sum formula

Consider the identity

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad (n \geq 1).$$

The notation on the left adds every integer from 1 through n . It can also be written $\sum_{i=1}^n i$, where i is a temporary summation index.

Worked example. For $n = 1$, both sides equal 1. Assume for an arbitrary $k \geq 1$ that $1 + \cdots + k = k(k+1)/2$. The next sum contains all the old terms plus one new term:

$$\begin{aligned} 1 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

The first equality is where the induction hypothesis is used. The final expression is exactly the claimed formula with n replaced by $k+1$.

Practice. Prove $1 + 3 + 5 + \cdots + (2n-1) = n^2$ for $n \geq 1$.

Solution. The base case is $1 = 1^2$. Assume the sum of the first k odd positive integers is k^2 . The next odd number is $2(k+1) - 1 = 2k+1$, so the next sum is

$$k^2 + (2k+1) = (k+1)^2.$$

This proves the inductive step and hence the identity for all positive n . The calculation is short because adding the next odd number supplies exactly the amount needed to grow a square of side k to one of side $k+1$.

Common error. Keep the “next term” separate from the old sum before substituting the hypothesis. Replacing the entire $(k+1)$ -term sum by the unproved formula would assume the desired result.

39 Induction for divisibility

Divisibility problems often become manageable when the next expression is rewritten as a multiple of the previous expression plus a known divisible quantity. This creates a place to use the induction hypothesis.

Worked example. Prove $3 \mid (4^n - 1)$ for every integer $n \geq 0$. At $n = 0$, the expression is 0, which is divisible by 3. Assume $4^k - 1 = 3m$ for some integer m . Then

$$\begin{aligned} 4^{k+1} - 1 &= 4(4^k - 1) + 3 \\ &= 4(3m) + 3 = 3(4m + 1). \end{aligned}$$

Since $4m + 1$ is an integer, the next expression is divisible by 3. This proves the step.

The useful decomposition was not $4 \cdot 4^k - 1$ alone. Adding and subtracting the appropriate constant exposed the expression $4^k - 1$ that the hypothesis already controls.

Practice. Prove $6 \mid (7^n - 1)$ for every integer $n \geq 0$.

Solution. At $n = 0$, $7^0 - 1 = 0$. Assume $7^k - 1 = 6m$. Then

$$\begin{aligned} 7^{k+1} - 1 &= 7(7^k - 1) + 6 \\ &= 7 \cdot 6m + 6 \\ &= 6(7m + 1). \end{aligned}$$

Thus the next value is also divisible by 6. The argument establishes the result for every nonnegative integer by induction.

Common error. Saying “the expression is clearly divisible” hides the key step. Identify an integer multiplier or cite a previously justified divisibility rule. If using congruence notation, define $a \equiv b \pmod{m}$ to mean that m divides $a - b$; it is not the same use of $\equiv$ as logical equivalence, though the context distinguishes them.

After learning the inductive proof, notice the shorter modular explanation $7 \equiv 1 \pmod{6}$. Different methods can establish the same fact; the inductive version specifically practises how a property is preserved from one index to the next.

40 Induction for inequalities

An inductive inequality proof must preserve the direction of every comparison. Multiplication by a positive quantity preserves the direction; multiplication by a negative quantity reverses it. The starting index is often determined by where an auxiliary inequality becomes valid.

Worked example. Prove $2^n \geq n + 1$ for $n \geq 0$. At $n = 0$, both sides equal 1. Assume $2^k \geq k + 1$. Multiplying by 2 gives

$$2^{k+1} \geq 2(k + 1) \geq k + 2,$$

where the second inequality holds because $k \geq 0$. The right side is $(k + 1) + 1$, so the desired next case follows.

Practice. Prove $2^n \geq n^2$ for all integers $n \geq 4$.

Solution. At $n = 4$, both sides equal 16. Assume $2^k \geq k^2$ for some $k \geq 4$. Then $2^{k+1} \geq 2k^2$. We need $2k^2 \geq (k + 1)^2$, which is equivalent to

$$k^2 - 2k - 1 \geq 0.$$

For $k \geq 4$, $k^2 - 2k - 1 = (k - 1)^2 - 2 \geq 9 - 2 > 0$. Therefore $2^{k+1} \geq (k + 1)^2$, completing the step.

Common error. The inequality is false at $n = 3$, since $8 < 9$. Starting at $n = 0$ and checking only that the initial equality holds would not solve the problem: the intended successor step must also work at every intermediate index. Induction cannot jump over a failed case without an explicit change in its structure.

When planning, write the target expression for $k + 1$ before manipulating the hypothesis. This makes clear which numerical comparison must bridge the old bound and the new one.

41 Strong induction

Strong induction permits the step for $k + 1$ to use all earlier established cases $P(n_0), \dots, P(k)$, rather than only $P(k)$. It proves the same kinds of statements as ordinary induction, but its hypothesis is better suited to problems that split an object into smaller pieces of unpredictable sizes.

Worked example. Every integer $n \geq 2$ is a product of primes. The base case 2 is prime. Assume every integer from 2 through k has a prime factorisation. For $k + 1$, if it is prime, it already has a one-factor prime representation. If it is composite, write $k + 1 = ab$ with $2 \leq a, b < k + 1$. Both factors lie among the earlier cases, so each is a product of primes. Multiplying those representations gives a prime factorisation of $k + 1$.

The argument establishes existence of a factorisation. It does not prove uniqueness; that is a separate theorem.

Practice. Suppose a sequence is defined by $a_0 = 0$, $a_1 = 1$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$. Prove $a_n \geq 0$ for every $n \geq 0$.

Solution. The first two values are nonnegative. For $n \geq 2$, assume every earlier value is nonnegative. In particular, $a_{n-1} \geq 0$ and $a_{n-2} \geq 0$. Their sum is non-negative, so $a_n \geq 0$. Strong induction completes the proof. A two-step induction hypothesis would also be sufficient, since the recurrence only uses two earlier terms.

Common error. Verify enough base cases for every earlier index used by the step. A recurrence involving a_{n-2} cannot be applied at $n = 1$ unless that term is separately defined.

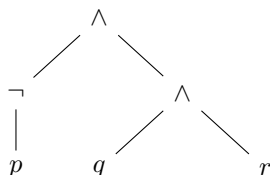
42 Induction diagrams and recursive structures

Induction is not limited to numerical formulas. A recursively defined object is built from smaller objects according to construction rules. *Structural induction* proves a property for the initial objects and shows that each allowed construction preserves it.

Worked example. Consider propositional formulas built from atoms using only negation and conjunction. Let $a(F)$ count atom occurrences and let $b(F)$ count conjunction symbols in a formula F . Then

$$a(F) = b(F) + 1.$$

For an atom, $a = 1$ and $b = 0$. If the identity holds for G , adding a negation changes neither count, so it holds for $\neg G$. If it holds for G and H , then $G \wedge H$ has $a(G) + a(H)$ atom occurrences and $b(G) + b(H) + 1$ conjunctions. Substitution shows that the atom count is again one greater.



This tree represents $(\neg p) \wedge (q \wedge r)$: it has three atom occurrences and two conjunctions.

Practice. Does the same count remain valid if disjunction is also allowed but $b(F)$ still counts only conjunctions?

Solution. No. The formula $p \vee q$ has two atom occurrences and zero conjunctions. The repaired statement counts *all binary connectives*, including disjunction. Every binary construction joins two formula trees and adds one binary node, preserving the corrected identity.

Common error. A structural proof must cover every constructor in the definition. Proving the conjunction case while forgetting negation leaves a gap even if the missing case is easy.

43 Diagnosing faulty induction arguments

An induction proof can fail through a false base case, an invalid algebraic step, a hidden restriction, or an induction hypothesis that does not actually support the next case. A particularly instructive failure occurs when the first transition behaves differently from later ones.

Worked example. Consider the false claim that every finite group of horses has the same colour. A proposed proof removes one horse, applies induction to the remaining k horses, then removes a different horse and repeats. It tries to connect the two groups through a horse in their overlap. When passing from one horse to two, however, the two one-horse groups have no common member. The connecting argument therefore fails exactly at the first needed transition.

This is a logical defect, not a disagreement about horses. The inductive step was not proved for every k in the claimed range.

Practice. A student wants to prove $1 + \cdots + n = n^2$. They write, “Assume it for k . Then the next sum is $k^2 + k + 1 = (k + 1)^2$.” Find all essential errors.

Solution. The asserted equality is false: $(k + 1)^2 = k^2 + 2k + 1$, not $k^2 + k + 1$. The claimed formula itself fails at $n = 2$, where the sum is 3 rather than 4. Although the base case $n = 1$ works, the incorrect algebra prevents a valid step. The induction hypothesis may be used temporarily to prove an implication, but it does not permit a false arithmetic equality.

Common error. A proof can look formally complete while containing one unsupported bridge. Read it as a chain of implications and ask, at every line, exactly which assumption or identity justifies the move.

For your own proofs, check the smallest nontrivial transition explicitly before polishing the general algebra. This often reveals missing base cases or hidden dependence on the size of an overlap.

44 Choosing a proof technique

The logical shape of a claim suggests a starting method. For a universal conditional, assume an arbitrary object satisfies the hypothesis. For existence, try to construct a witness. For uniqueness, compare two possible witnesses. For a sequence of assertions indexed by integers, look for an inductive relation.

Use contraposition when a negated conclusion is easy to describe. Use contradiction when a denied claim creates a useful object, such as a smallest counterexample. Use cases when the domain naturally separates by parity, sign, or remainder. These are choices guided by structure, not rigid rules.

Worked example. Claim: if the product of two integers is odd, both integers are odd. The contrapositive is especially natural: if at least one factor is even, the product is even. If $a = 2k$, then $ab = 2(kb)$; the same reasoning applies if b is even. This proves the contrapositive and hence the claim.

Practice. Select and carry out a short method for: (a) there exists an even prime; (b) no integer is both even and odd; (c) if $a \mid b$, then $a \mid bc$ for integer c .

Solution. (a) A witness is 2: it is even and has exactly two positive divisors, 1 and 2. (b) Suppose $n = 2r = 2s + 1$. Then $2(r - s) = 1$, impossible for integers because an even integer cannot equal 1; equivalently, $r - s = 1/2$ contradicts integrality. (c) Directly, write $b = ak$. Then $bc = a(kc)$, and kc is an integer.

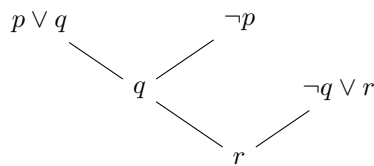
Common error. Do not choose induction merely because a statement mentions integers. If its proof does not depend on moving from smaller values to larger ones, a direct argument may be simpler and more informative.

45 Propositional resolution and refutation

Resolution is an inference rule for clauses. From $A \vee p$ and $B \vee \neg p$, infer $A \vee B$. Here A and B denote disjunctions of any remaining literals, possibly none. The rule eliminates a pair of complementary literals from two different clauses.

Why is it sound? If the conclusion were false, both A and B would be false. The first premise would then require p , while the second would require $\neg p$, an impossibility. Thus true premises cannot produce a false resolvent.

Worked example. Prove r from $p \vee q$, $\neg p$, and $\neg q \vee r$ by refutation. Add $\neg r$, the negation of the conclusion. Resolve $p \vee q$ with $\neg p$ to obtain q . Resolve q with $\neg q \vee r$ to obtain r . Finally resolve r with $\neg r$ to obtain the *empty clause*, written $\square$, which represents falsehood.



Practice. Resolve $p \vee q$ with $\neg p \vee s$. Is the result $q \wedge s$?

Solution. The resolvent is $q \vee s$. At least one remaining alternative must hold; both are not required. For example, $p = \text{F}, q = \text{T}, s = \text{F}$ satisfies both premises but falsifies $q \wedge s$.

Common error. Resolution proves an entailment by showing that premises plus a negated conclusion are inconsistent. Failing to find an empty clause in a few steps is not itself proof that the premises are satisfiable.

46 Substitution and unification in predicate logic

Predicate resolution must make complementary atoms match. A *substitution* assigns terms to variables, for example $\theta = \{x \mapsto a, y \mapsto f(a)\}$. Applying it replaces every relevant free occurrence consistently.

Unification finds a substitution that makes two expressions identical. For $P(x, b)$ and $P(a, y)$, the substitution $\{x \mapsto a, y \mapsto b\}$ works. A *most general unifier* imposes only the bindings necessary to achieve the match, leaving other variables unrestricted.

Worked example. From $\neg S(x) \vee P(x)$ and $S(\text{Mira})$, unify $S(x)$ with $S(\text{Mira})$ using $x \mapsto \text{Mira}$. Apply this substitution to the entire first clause, giving $\neg S(\text{Mira}) \vee P(\text{Mira})$. Resolution yields $P(\text{Mira})$.

Repeated variable occurrences are constraints. The atoms $P(x, x)$ and $P(a, b)$ can unify only if the terms a and b can themselves be made identical. Distinct constant symbols do not syntactically unify, even though ordinary first-order semantics does not automatically require them to denote different objects.

Practice. Unify $R(x, f(y))$ and $R(f(a), f(b))$. Then explain why x and $f(x)$ do not unify as finite first-order terms.

Solution. The first pair has unifier $\{x \mapsto f(a), y \mapsto b\}$. For the second pair, assigning $x = f(x)$ would require x to contain itself as a proper subterm, leading to an infinite expansion. The *occurs check* rejects such a substitution in standard finite-term unification.

Common error. Before resolving clauses, rename their variables apart when needed. Variables in separate universally quantified clauses are independently bound; identical printed names should not accidentally force an unintended relationship.

47 Preparing predicate formulas for resolution

Predicate resolution works with clauses whose remaining variables are treated as universally quantified. Conversion begins by removing implications, moving negations inward, and renaming bound variables to avoid collisions. Existential quantifiers require an additional operation: *Skolemisation*.

An existential witness outside the scope of universal quantifiers can be represented by a fresh constant. A witness that may depend on preceding universally quantified variables needs a fresh function of those variables.

Worked example. The formula $\forall x \exists y R(x, y)$ is Skolemised to $\forall x R(x, f(x))$. The function f records a potentially different witness for each x . In contrast, $\exists y \forall x R(x, y)$ uses one constant c , giving $\forall x R(x, c)$.

Skolemisation preserves satisfiability in an expanded language; it is not ordinary logical equivalence in the original vocabulary. If the original formula has a model, its witnesses can interpret the new symbols. Conversely, forgetting those new symbols recovers a model of the original existential claim.

Practice. Convert $\forall x (P(x) \rightarrow \exists y Q(x, y))$ to a clause suitable for resolution.

Solution. Remove the implication to obtain $\forall x (\neg P(x) \vee \exists y Q(x, y))$. Since y is not free in $P(x)$, the existential quantifier can be moved to cover the disjunction. Its witness may depend on x , so introduce $f(x)$. The resulting clause is

$$\neg P(x) \vee Q(x, f(x)),$$

with x understood as universally quantified.

Common error. Using a single constant instead of $f(x)$ would require a common witness for all x , which the original claim does not require. Also keep fresh Skolem symbols separate from existing constants and functions; they encode new witness choices.

48 Answer extraction from a resolution proof

An ordinary proof of $\exists x Q(x)$ establishes existence. *Answer extraction* also records which substitutions make $Q(x)$ follow from the knowledge base. In this section, the syllabus term refers to extracting answer terms from resolution reasoning.

Suppose the facts and rule are

$$S(\text{Mira}), \quad \forall x(S(x) \rightarrow E(x)),$$

where S means registered and E means eligible. The query asks for an x satisfying $E(x)$. Introduce a fresh predicate Answer and the query clause

$$\neg E(x) \vee \text{Answer}(x).$$

The answer literal carries the requested variable through the derivation. It does not impose a new factual property on the original domain.

Worked example. Convert the rule to $\neg S(y) \vee E(y)$, using a separate variable name. Resolve it with the query clause, matching $E(x)$ and $E(y)$, to get

$$\neg S(y) \vee \text{Answer}(y).$$

Resolve with $S(\text{Mira})$, binding y to Mira . The result is $\text{Answer}(\text{Mira})$. Removing the bookkeeping predicate reports Mira as an answer and the substitution $x \mapsto \text{Mira}$ as its explanation.

Practice. Add $S(\text{Arun})$. What answers can be extracted?

Solution. Resolving the remaining registration literal against the first fact yields Mira ; resolving it against the new fact yields Arun . These are separate successful derivations. Both answers are justified by the same rule. The procedure should continue searching if all answers are requested rather than stopping after its first success.

Common error. A witness chosen to interpret an existential assumption is not automatically a uniquely identified real-world individual. Report the term or substitution actually supported by the proof, and distinguish a named answer from an existence guarantee.

49 Answer extraction through a chain of relations

A query can require several related facts. The crucial discipline is to keep shared variables consistent as each subgoal is satisfied. A later match must respect earlier substitutions.

Worked example. Let $P(a, b)$ mean that a is a parent of b , and let

$$P(\text{Asha}, \text{Bina}), \quad P(\text{Bina}, \text{Chirag})$$

be facts. The rule for a grandparent is

$$P(x, y) \wedge P(y, z) \rightarrow G(x, z).$$

To answer “Who is a grandparent of Chirag?”, use the clause $\neg G(w, \text{Chirag}) \vee \text{Answer}(w)$. Resolve it with the rule’s clause to obtain

$$\neg P(x, y) \vee \neg P(y, \text{Chirag}) \vee \text{Answer}(x).$$

Matching the second parent condition with $P(\text{Bina}, \text{Chirag})$ binds y to Bina. The remaining clause is $\neg P(x, \text{Bina}) \vee \text{Answer}(x)$. Matching the first fact binds x to Asha and yields $\text{Answer}(\text{Asha})$.

Practice. If $P(\text{Dev}, \text{Bina})$ is also known, what changes? Does failure to find another answer prove that no other grandparent exists?

Solution. Dev becomes another derived answer, through the same intermediate person Bina. Failure to derive further answers does not by itself establish their real-world nonexistence. Under ordinary first-order, open-world reasoning, the knowledge base may simply omit relevant facts. A complete search can characterise consequences of a specified finite rule system, but an additional closed-world convention is needed to treat missing facts as false.

Common error. In general first-order resolution, search may not terminate, and answers can contain variables or disjunctions rather than a single ground name. The examples here use definite rules and named facts so that each successful derivation has an easily interpretable answer substitution.

50 Integrated revision and notation reference

The unit has three connected levels. Propositional logic treats statements as indivisible truth-valued objects. Predicate logic describes objects and relations inside statements. Proof methods organise valid reasoning so that it covers all required cases. Resolution provides a clause-based route to entailment and, with recorded substitutions, to query answers.

Notation	Meaning
$\neg p$	not p
$p \wedge q, p \vee q$	and; inclusive or
$p \rightarrow q, p \leftrightarrow q$	conditional; biconditional
$A \equiv B$	logical equivalence
$\Gamma \models C$	premise set Γ entails C
$\Gamma \vdash C$	C is derivable in a proof system
$\forall x, \exists x$	for every x ; for some x
θ	a substitution
$\square$	empty clause, representing falsehood
$a \mid b$	b is an integer multiple of a

The distinction between $\models$ and $\vdash$ is semantic versus syntactic. The first concerns truth in all models; the second concerns an available derivation. Soundness of a proof system says that derivability guarantees entailment. Completeness says that every entailment in the relevant class can be derived, subject to the precise system and assumptions.

Practice. Explain why $\Gamma \models C$ is equivalent to unsatisfiability of $\Gamma \cup \{\neg C\}$.

Solution. Entailment rules out a model where every premise is true and C is false. Such a model would be exactly a model of all premises together with $\neg C$. Saying that no such model exists is the definition of unsatisfiability. This is why refutation methods add the negated conclusion instead of trying to guess the conclusion directly.

When revising, practise moving between words, formulas, models, and proofs. Knowing only a symbolic recipe makes translation errors difficult to detect; knowing only an intuitive explanation makes long arguments hard to verify.

51 Cumulative practice questions

Attempt these questions without the solutions in the next section. For every symbolic answer, include a sentence explaining what it means. For a false universal claim or invalid argument, provide a specific counterexample or countermodel.

1. Simplify $\neg(p \rightarrow q) \vee (p \wedge q)$.
2. Find a CNF equivalent to $(p \wedge q) \rightarrow r$. Is it a tautology?
3. Decide whether $p \rightarrow q$, $q \rightarrow r$, and $\neg r$ entail $\neg p$. Give a short derivation.
4. Over integers, compare the truth of $\forall x \exists y (x + y = 0)$ and $\exists y \forall x (x + y = 0)$.
5. Negate “Every student has read some book” using separate domains of students and books. Explain the resulting sentence.
6. Translate “Exactly one student solved every problem.” Use $S(x)$ for student, $R(y)$ for problem, and $L(x, y)$ for solved, with a common domain of relevant objects.
7. Prove that the square of an odd integer leaves remainder 1 on division by 8.
8. Prove by induction that $1 + 2 + 4 + \cdots + 2^n = 2^{n+1} - 1$ for $n \geq 0$.
9. Use resolution to show that $p \vee q$, $\neg p \vee r$, and $\neg q \vee r$ entail r .
10. Given $A(a)$, $\forall x (A(x) \rightarrow B(x))$, and $\forall x (B(x) \rightarrow C(x))$, extract an answer to the query $C(z)$, showing the bindings.

Suggested working method. For Questions 1–3 and 9, identify the connectives before applying a law or rule. For Questions 4–6, state the permitted dependence of each witness. For Questions 7–8, write the domain, the definition or induction hypothesis, and the precise final form needed. For Question 10, carry the answer variable through the proof instead of simply stating that an answer exists.

Self-check. A complete solution should reveal why it works. The final formula alone is insufficient for a proof question, while several pages of calculation are unnecessary for a counterexample question once the relevant failure is explicitly demonstrated.

52 Cumulative solutions and further reading

Solutions 1–5

1. Since $\neg(p \rightarrow q) \equiv p \wedge \neg q$, the expression becomes $(p \wedge \neg q) \vee (p \wedge q) \equiv p \wedge (\neg q \vee q) \equiv p$.
2. Removing the implication and applying De Morgan's law gives $\neg p \vee \neg q \vee r$. This is a single CNF clause. It is not a tautology: $p = q = \text{T}, r = \text{F}$ makes it false. It is satisfiable, for example whenever $r = \text{T}$.
3. From $q \rightarrow r$ and $\neg r$, modus tollens gives $\neg q$. From $p \rightarrow q$ and $\neg q$, another use of modus tollens gives $\neg p$. Thus the argument is valid.
4. The first statement is true: for each integer x , choose the integer $y = -x$. The second is false: $x = 0$ would require $y = 0$, while $x = 1$ would require $y = -1$. One fixed y cannot meet both demands.
5. If $L(x, y)$ means that student x read book y , the original is $\forall x \exists y L(x, y)$. Its negation is $\exists x \forall y \neg L(x, y)$: some student has read no book in the stated domain.

Solutions 6–10

6. Define $H(x)$ as an abbreviation for $S(x) \wedge \forall y(R(y) \rightarrow L(x, y))$. Then write $\exists x(H(x) \wedge \forall z(H(z) \rightarrow z = x))$. This permits other students to solve some problems, but exactly one solves every problem. The abbreviation is fully specified and can be expanded if required.
7. Write $n = 2k + 1$. Then $n^2 = 4k(k + 1) + 1$. One of the consecutive integers $k, k + 1$ is even, so $k(k + 1) = 2m$ for some integer m . Hence $n^2 = 8m + 1$, proving the remainder statement.
8. At $n = 0$, the sum is $1 = 2^1 - 1$. Assume the sum through 2^k equals $2^{k+1} - 1$. Adding the next term gives $(2^{k+1} - 1) + 2^{k+1} = 2^{k+2} - 1$, the required case at $k + 1$. Therefore induction applies for all $n \geq 0$.
9. Add $\neg r$. Resolving it with $\neg p \vee r$ yields $\neg p$, and with $\neg q \vee r$ yields $\neg q$. Resolve $p \vee q$ with $\neg p$ to get q , then with $\neg q$ to get $\square$. Thus premises plus the negated conclusion are unsatisfiable.
10. Start with $\neg C(z) \vee \text{Answer}(z)$. Resolve against $\neg B(y) \vee C(y)$ to obtain $\neg B(y) \vee \text{Answer}(y)$. Resolve against $\neg A(x) \vee B(x)$ to obtain $\neg A(x) \vee \text{Answer}(x)$. The fact $A(a)$ binds the remaining variable to a , giving $\text{Answer}(a)$. The answer substitution is $z \mapsto a$.

Further reading and scope

The explanations and exercises in these notes are original instructional material. For additional treatments, consult Eric Lehman, F. Thomson Leighton, and Albert R. Meyer, *Mathematics for Computer Science*, available through MIT OpenCourseWare. Its logic and proof chapters provide a broader discrete-mathematics context.

For the resolution-based meaning of answer extraction, see the Stanford Introduction to Logic supplement on resolution and the University of Texas notes on answer extraction. Here answer extraction is developed with explicit substitutions and definite-rule examples; general automated theorem proving has additional search and completeness issues.